

Ankit Singh

Curriculum Vitae

Research Interests: Computer Vision, Real-Time Perception, Edge AI, Human-AI Collaboration, Scalable Intelligent Systems, Full Stack AI Application

Contact

Email: ankitsingh388667@gmail.com, LinkedIn: linkedin.com/in/ankit-singh-5a481b2a3

GitHub: github.com/AnkitSingh10-hub, GitLab: gitlab.com/ankitsingh388667

Education

Pokhara University (NCIT)

Bachelor of Engineering in Software Engineering

Sep 2018 - Aug 2023

Nepal

- GPA: 3.52/4.0
- English Proficiency: TOEFL iBT 99/120

Relevant Coursework

- **Programming Languages (PL):** Principles of Programming Languages, Data Structures & Algorithms, Analysis & Design of Algorithms, Mathematical Foundation of Computer Science
- **Software Engineering (SE):** Software Testing, Verification, Validation & Quality, Applied Operating Systems, Database Management Systems, Web Technology
- **Machine Learning (ML):** Artificial Intelligence and Neural Networks, Probability & Queuing Theory, Numerical Methods, Engineering Mathematics (I, II, III)

Selected Research Initiatives and Projects

DermaEval: Adaptive Intelligent Tutoring System (ITS) for Dermatology

Jan 2026 – Present

- **Tech Stack:** Vue.js (Konva), Django REST, Celery, Redis, PyTorch (YOLOv11), Active Learning Logic
- **Adaptive Curriculum Algorithm:** Engineered an **Active Learning** engine that analyzes trainee "Confusion Matrices" (e.g., consistent misclassification of Rosacea vs. Acne). The system dynamically queries the dataset to serve targeted image fidelities that address specific latent knowledge gaps.
- **Diagnostic Reasoning Capture:** Designed a **Human-in-the-Loop** interface to capture not just bounding boxes, but semantic feature tags (e.g., "Irregular Border"). This creates a dense dataset linking visual saliency to human diagnostic error, enabling research into cognitive load and visual search patterns.
- **Asynchronous Evaluation Pipeline:** Decoupled heavy computer vision inference using **Celery** and **Redis**, allowing for real-time grading of complex annotations against expert Ground Truths (IoU) without inducing UI latency.

SurgiCount: Computer Vision for Surgical Safety & Uncertainty Quantification

Dec 2025 – Jan 2026

- **Tech Stack:** Vue.js, Django REST, Docker, YOLOv11, OpenCV, Synthetic Data Augmentation
- **Robust Vision with Synthetic Data:** Developed a safety-critical detection pipeline using **YOLOv11**. Utilized **Synthetic Data Generation** to augment training sets with simulated blood, glare, and occlusion, significantly improving model robustness against Retained Surgical Bodies (RSB) in noisy environments.
- **Uncertainty Quantification (UQ):** Engineered a "Confidence-Aware" protocol where low-probability detections trigger a mandatory **Human-in-the-Loop** verification. This explicitly models the "Hand-off" between AI and human nurses, preventing false negatives in high-stakes scenarios.

- **State-Aware Audit Trail:** Implemented a strict **Django** state machine to enforce "Check-in/Check-out" logic. The system logs "Near-Miss" events—instances where AI flagged discrepancies that humans initially missed—providing quantitative data on the efficacy of AI-assisted safety checks.

Industry Experience

Pathway Technologies and Services
Software Engineer | Research Data Infrastructure Lead

Nov 2023 – Present
Kathmandu, Nepal

Focus 1: Research Engineering & Data Infrastructure

- * **Research Pipeline Architecture:** Architected the VCC (Virtual Collection Center) platform serving as the technical backbone for a multi-year Randomized Controlled Trial (RCT) led by UC Davis economists. The system analyzes how reducing information frictions facilitates market linkages for 11,700+ farmers.
- * **Socio-Technical Impact:** Built the Agri-Input module as a **Decision Support System (DSS)**, optimizing supply chain logistics. Enabled researchers to correlate **access to quality inputs** with **quantifiable gains in farmer productivity**.
- * **High-Fidelity Dataset Generation:** Engineered a location-based query routing system (Village Animal Health Worker), pairing farmers with veterinary experts. Incorporates Gemini APIs to assist response generation, functioning as a human-in-the-loop pipeline to capture expert-verified ground truth for NLP tasks.
- * **Data Provenance & Quality:** Designed strict data validation protocols to ensure high-fidelity inputs from **11,700+ farmers and 15,000+ livestock**, preserving statistical integrity for academic impact modeling.

Focus 2: Software Architecture & Scalable Systems

- * **Offline-First Distributed System:** Engineered a robust synchronization engine (Django REST/PostgreSQL) to handle **data sparsity and latency**, ensuring 100% data recoverability in remote regions.
- * **High-Concurrency Orchestration:** Built asynchronous workflows using **Celery and Redis** to process high-volume transactional events (100k+ NPR value), ensuring zero-latency UI performance.
- * **FinTech Integration:** Developed a ledger-based commerce engine using atomic transactions for distributed districts, handling automated reconciliation via eSewa/Pathao/Magnus APIs.
- * **Deployment at Scale:** Orchestrated the rollout of "Geokrishi Sewa" across 77 cooperatives with Heifer International, ensuring data recoverability via offline-first architecture.

Awards & Honors

Associate AI Engineer, Datacamp	2025
Certificate of Appreciation, Pathway Technologies & Heifer International	2025
KINAM Prototype Certification, Delivered functional prototype within semester timeline	2022

Mentorship & Knowledge Transfer

Leadership & Capacity Building

- * **Technical Training:** Spearheaded internal workshops on Django REST, Docker, and Database Design, guiding junior developers through code reviews.
- * **Implementation Support:** Conducted technical training for 70+ cooperative managers and extension workers in remote Nepal, ensuring adherence to data collection protocols.

Skills

Technical Skills

Languages: Python, C/C++, JavaScript/TypeScript, Java, SQL

Web Frameworks: Django (REST), Vue.js, Celery, Redis, Git

AI/ML: PyTorch, YOLOv11, ResNet, Scikit-Learn, OpenCV, Albumentations

ML Ops: Roboflow, Docker, CI/CD Automation, AWS, PostgreSQL

Soft Skills

Adaptability, Proactive, Leadership, Communication, Time Management, Problem Solving